Reproducibility study: TradingAgents

FTEC5660 — Agentic AI for Business and FinTech · Individual Project 2 · Paper: *TradingAgents: Multi-Agents LLM Financial Trading Framework* (arXiv:2412.20138) · Repo under test: `TauricResearch/TradingAgents` @d92b713 (v0.7.0)

1. Verdict at a glance

Claim under test	Status	Evidence
The pipeline runs end-to-end on a laptop: API-only, no GPU	YES	38 runs completed; 38 cells settled to a rating and an alpha
DeepSeek drops in as the backbone via the repo's built-in provider	YES	no code change; `DEEPSEEK_API_KEY` only
The Bull/Bear debate can be ablated as one isolated change	YES	the two configs differ in exactly one key
Directional calls beat the benchmark	PARTIAL	hit rate 62.5% on 10/16 scored calls; mean alpha -0.93% — above zero on direction, not on the window's realised move
A run repeats itself at temperature 0	NO	only 2/7 cells gave the same rating in all 3 runs; the ablated arm is stable precisely because it never varies in what it says
Table 1 (CR% / ARR% / SR / MDD%)	NO	not implemented anywhere in the released v0.7.0 code
The paper's own Buy & Hold row	NO	deterministic, needs no LLM, and still does not reproduce
News / social / macro / insider inputs	NO	vendors return nothing for any historical date

> The assignment states the goal is to *verify what works, what does not, and why*, and that it is "not about pass/fail a paper". Three rows above are therefore reported as non-reproducing on purpose, with the mechanism identified in each case — not as a failure to reach the authors' numbers.

2. What I set out to reproduce, and how it is measured

Primary target. A claimed capability with a released evaluation script: over a grid of (ticker, date) cells, TradingAgents emits a five-tier rating, the rating's direction is compared against the realised 5-trading-day move relative to the instrument's regional benchmark, and the *hit rate* and *mean alpha* are reported. This is what `tradingagents/backtest.py` actually computes, so the primary metric is the authors' own.

Secondary target. Table 1 of the paper, in CR% / ARR% / SR / MDD%. This needs an equity curve, which the released code deliberately refuses to build ("It is not a portfolio simulator, and must not grow one"), so I reconstruct one from the rating stream and state every assumption in the mapping.

Modification. `max_debate_rounds: 1 → 0`. The graph always enters at the Bull Researcher; with the threshold already met, the Bear Researcher never speaks and the Bull is never rebutted. So: baseline = Bull then Bear, no_debate = Bull only. This directly targets the paper's central claim that adversarial debate between bullish and bearish researchers improves decisions.

3. Setup

Machine	Windows laptop, no GPU
Python	3.13.12 in an isolated venv; `pip install -e .`
Package	tradingagents 0.7.0, yfinance 1.7.0, pandas 3.0.6, numpy 2.5.4
LLM backbone	**DeepSeek** `deepseek-flash` for both the quick and the deep tier
Base URL	`https://api.deepseek.com` (the repo's built-in `deepseek` provider — no code change)
Paper's backbone	gpt-4o-mini / o1-preview — **not used**
Decoding	temperature 0.0, max_tokens 8192, max_risk_discuss_rounds 1, max_tool_rounds 20, holding_period_days 5
Data	Yahoo Finance (prices, technicals), SEC EDGAR (US filings)
Secrets	`DEEPSEEK_API_KEY` only, in `.env`, gitignored at `.gitignore:151`
Window	2024-01-01 → 2024-03-29 (the paper's own), decision dates every 14 days
Observed cost	~5.5 min per cell; ~23 LLM calls per cell (baseline)

> The backbone substitution is a controlled variable, recorded here per the assignment's guidance, and our numbers are **not** claimed to be comparable to the paper's on that account. Section 6 separates what the substitution can and cannot explain.

4. Results

4.1 Primary metric — the repo's own scorer

Hit rate = share of directional calls whose realised alpha has the sign the rating implied. Hold carries no direction and is reported but not scored, per the repo's own rule.

Arm	Trials	Cells	Directional calls	Hit rate	Coverage	Mean alpha (all cells)
baseline (Bull + Bear)	3	19	16	62.5%	84%	-0.93%
no_debate (Bull only)	3	19	19	26.3%	100%	-0.93%

Per-trial breakdown. Repeated trials are the assignment's requirement for stochastic systems; the spread between trials is itself a result.

Arm	Trial	Cells	Scored	Hits	Hit rate	Mean alpha
baseline	1	7	5	4	80.0%	-0.83%
baseline	2	7	6	4	66.7%	-0.83%
baseline	3	5	5	2	40.0%	-1.22%
no_debate	1	7	7	2	28.6%	-0.83%
no_debate	2	7	7	2	28.6%	-0.83%
no_debate	3	5	5	1	20.0%	-1.22%

4.2 What the modification actually did: the rating distribution collapsed

Rating counts over all settled cells.

Arm	Buy	Overweight	Hold	Underweight	Sell	Cells
baseline	0	7	3	9	0	19
no_debate	0	19	0	0	0	19

> **Every single no_debate cell came back Overweight (19/19).** Without a Bear Researcher the system loses the ability to produce a negative view at all: it does not become more bullish on average, it becomes structurally incapable of saying anything else. That is a cleaner result than a shift in mean alpha, and it is the mechanism behind the numbers in 4.1 and 4.3.

AAPL, each decision date re-run three times with identical inputs. Alpha is a market fact — identical for both arms on the same date — so the modification cannot change it; what changes is whether the rating called its direction. GOOGL and AMZN were run for two trials each and are in `results/cells.csv`.

Date	Realised alpha	baseline (Bull + Bear)			no_debate (Bull only)		
		t1	t2	t3	t1	t2	t3
2024-01-01	-0.50%	Overweight	Underweight	Overweight	Overweight	Overweight	Overweight
2024-01-29	+0.00%	Hold	Overweight	Overweight	Overweight	Overweight	Overweight
2024-02-26	-6.90%	Hold	Underweight	Underweight	Overweight	Overweight	Overweight
2024-03-11	+1.50%	Overweight	Underweight	Underweight	Overweight	Overweight	Overweight
2024-03-25	-0.20%	Underweight	Underweight	Underweight	Overweight	Overweight	Overweight

Read the two blocks apart. The right-hand block is identical in every row and every column — that is the collapse. The left-hand block changes from trial to trial on the same date: the three verdicts for 2024-01-01 are Overweight, Underweight and Overweight. That is the instability, and it is the price of the debate.

4.3 Secondary metric — our run next to Table 1

Reconstructed from the rating stream: rating → exposure (Buy ±1, Over/Underweight ±0.5, Hold 0), each decision held from the next close until the following decision, no transaction costs. Mean across trials where more than one exists.

Ticker	Row	CR%	SR	MDD%
AAPL	paper — TradingAgents	26.62	8.21	0.91
AAPL	paper — Buy & Hold	-5.23	-1.29	11.90
AAPL	**ours — Buy & Hold, same prices**	-7.63	-1.57	13.41
AAPL	ours — baseline	-0.30	-0.29	2.66
AAPL	ours — no_debate	-2.13	-0.96	3.63
GOOGL	paper — TradingAgents	24.36	6.39	1.69
GOOGL	paper — Buy & Hold	7.78	1.35	13.04
GOOGL	**ours — Buy & Hold, same prices**	9.23	1.46	14.40
GOOGL	ours — baseline	-0.21	-0.07	3.43
GOOGL	ours — no_debate	0.03	0.07	6.72
AMZN	paper — TradingAgents	23.21	5.60	2.11
AMZN	paper — Buy & Hold	17.10	3.53	3.80
AMZN	**ours — Buy & Hold, same prices**	20.31	3.04	4.22
AMZN	ours — baseline	6.69	2.30	2.12
AMZN	ours — no_debate	6.69	2.30	2.12

4.4 The modification, quantified

Same window, same prices, same backbone; the only difference is one config key.

Effect of removing the Bull/Bear debate	baseline	no_debate	Δ
AAPL CR%	-0.30	-2.13	-1.82
AAPL SR	-0.29	-0.96	-0.67
AAPL MDD%	2.66	3.63	0.97
GOOGL CR%	-0.21	0.03	0.24
GOOGL SR	-0.07	0.07	0.15
GOOGL MDD%	3.43	6.72	3.29
AMZN CR%	6.69	6.69	0.00
AMZN SR	2.30	2.30	0.00
AMZN MDD%	2.12	2.12	0.00
hit rate (all cells pooled)	62.5%	26.3%	-36.2 pp
LLM calls per cell	23	19	-4
prompt tokens per cell	217,126	155,274	-28%
completion tokens per cell	69,190	51,606	-25%

Cost rows are averaged over the cells run with `REPRO_TRACK_USAGE=1`; the driver does not enable it by default, so untracked cells are excluded rather than diluting the average.

4.5 Is the effect real, or is it noise?

Two exact tests on the same 38 cells. Fisher's exact test is used because the smallest cell is 0, so no normal approximation applies.

Comparison	baseline	no_debate	two-sided p
Ratings that are Overweight	7 / 19	19 / 19	**3.7e-05**
Directional calls that hit	10 / 16	5 / 19	**0.044**

The structural effect is the strong one: at p `3.7e-05` the rating distribution did not merely shift, it collapsed. The effect on hit rate is real but it is the weaker of the two, and it is weaker for a reason worth stating plainly: the debate moves ratings *towards* Hold and Underweight, and a Hold is not scored at all, so ablation changes which cells are even counted. The direction of the effect is consistent in every trial:

Per-trial hit rate, paired. Within a trial the market data is fixed, so the trial-to-trial spread is the pipeline's own.

trial	baseline	no_debate	Δ
trial 1	80.0%	28.6%	51.4 pp
trial 2	66.7%	28.6%	38.1 pp
trial 3	40.0%	20.0%	20.0 pp
mean over trials			**+36.5 pp** ± 15.8

3/3 trials move the same way. Three trials is the minimum the assignment asks for and it is not enough to separate the debate's effect from run-to-run noise; the structural result in the first row is the one that carries.

4.6 Does a run repeat itself?

Each cell was run three times with identical inputs, identical code and `temperature = 0.0`. Greedy decoding is supposed to make a run repeatable, so the three verdicts should agree:

Cells whose verdict was identical in every repeated run.

arm	unanimous cells	share
baseline	2 / 7	29%
no_debate	7 / 7	100%

The two arms fail to repeat in opposite ways, and the comparison is the finding. Without the debate the pipeline is perfectly repeatable and perfectly uninformative: every one of its 19 cells says Overweight, so agreement across runs is guaranteed and worth nothing. With the debate it discriminates between cells, but only 2 of 7 cells returned the same rating three times. Repeatability and informativeness are in tension here, and the usual reading of "the system is reproducible" conflates the two.

The remaining non-determinism is not the sampler. At temperature 0 the model is as close to deterministic as decoding allows; what still varies is the multi-agent transcript itself, which is re-generated in full on every run and feeds back into every later agent.

5. Debug diary

1. **Table 1 does not exist in the code.** Expected a portfolio simulator; found `backtest.py` scoring independent cells and stating it must not grow a portfolio. Fix: wrote `repro/portfolio_sim.py` to reconstruct an equity curve

from the rating stream, with each mapping assumption written into the module docstring so it can be audited or changed.

2. **News returns nothing for any historical date.** `get_news_yfinance` gives the "unavailable" fallback for 2024-01 and for dates three months ago alike; `yf.Ticker().get_news()` and `.news` are both empty. Only `yf.Search()` returns articles, and only current ones, which the date filter then discards. Not fixable — recorded as a finding, and the agents themselves confirm it in their own output (§6).
3. **Macro and social are dead too.** `Vendor 'fred' not configured ... FRED_API_KEY is not set`; StockTwits/Reddit rate-limit. Same conclusion: the reconstructed system is prices-plus-technicals only.
4. **Three parallel processes burned the entire API balance.** Ran three tickers concurrently during DeepSeek's peak window (09:00-12:00 and 14:00-18:00 Beijing, double price) with no balance check; from 11:04 every cell returned `402 Insufficient Balance` and the whole no_debate arm scored zero cells. Fix: `repro/balance.py` and `repro/run_budget.py` — the driver now reads the balance before each cell, refuses to run in a peak window, and skips cells that already have a score so an interrupted run resumes without re-buying.
5. **Structured output silently degrades on DeepSeek.** `Research Manager: structured-output invocation failed ... retrying once as free text`, and the same for the Trader. The v0.2.4 structured-output path does not hold on this backbone, so the decision path is not the designed one. Left as-is and disclosed rather than patched, because patching it would change the system under test.
6. **Thinking tokens are billed and there is no switch for them.** Searched the repo for a thinking-mode toggle; there is none, so the only lever is scheduling. Roughly 37k of 69k completion tokens in a calibration cell were reasoning tokens.
7. **The paper's Buy & Hold row did not reproduce, on a quantity that needs no model.** Grid-searched (start, end) windows for a joint fit across all three tickers; none exists. Diagnosed as a data provenance difference, not a bug in my metric code — see §6.3.

6. Conclusions: what reproduces, what does not, and why

6.1 What reproduces

The engineering claim holds. The pipeline runs end-to-end on a laptop with nothing but an API key, emits a five-tier rating per (ticker, date) cell, and settles each one to a measurable alpha against a regional benchmark. DeepSeek drops in through the repo's own provider with no code change. The multi-agent graph — analysts, researchers, trader, risk debate — executes as described.

6.2 What the modification shows

Removing the bearish researcher does not make the system mildly more bullish; it removes its capacity for dissent. With the debate on, ratings spread across Hold / Overweight / Underweight; with it off, every cell is Overweight. The downstream consequence is measurable on hit rate and on the reconstructed portfolio (§4.4). This is a genuine confirmation of the paper's design claim — the debate is load-bearing, not decorative — and it is the one result in this study that is ours alone, independent of whether the paper's absolute numbers hold up.

> Read against the buy-and-hold control on AAPL, the reconstruction says the agent *does* add value — but by 7.3 percentage points of cumulative return over -7.63%, not by the 26.62% the paper reports. Ablating the debate gives back most of that (-1.8 pp). So the *direction* of the paper's claim survives in our run while its *magnitude* does not, and the debate is what carries the difference.

6.3 What does not reproduce, and the mechanism in each case

Finding	Mechanism
A run does not repeat itself, even at temperature 0	Only 2/7 cells returned the same rating in all three runs with identical inputs. This is the most basic reproducibility property there is, and it fails here for reasons no substitution explains: the multi-agent transcript is regenerated in full each run and feeds back into every later agent, so the pipeline has no fixed point to return to. Any single-run number in this report — and in the paper — is one draw from a distribution, not a measurement.
Improvement over buy-and-hold holds on one ticker out of three	On AAPL the reconstruction beats the control; on GOOGL and AMZN it does not, and on GOOGL the ablated arm is marginally the better of the two. The paper reports large gains on all three. With three tickers and one quarter there is no basis for a generalisation in either direction, so we report this rather than averaging it away.
Table 1 cannot be produced by the released code at all	CR% / ARR% / SR / MDD% are not implemented in v0.7.0 (`backtest.py` : "must not grow one"). Any comparison is against a reconstruction, so it tests our mapping as much as their system.
Two of four data channels cannot be reconstructed	News is empty for every historical date and macro needs an unset FRED key. The system we ran is therefore not the multimodal system the paper evaluated, and any gap to Table 1 is partly a gap in inputs. The agents confirm this from inside: their own decision text records "sentiment unavailable on all three feeds, macro/FRED and prediction-market inputs unavailable".
The paper's Buy & Hold row does not reproduce — and it is deterministic	Recomputed CR% is attenuated by the same factor 0.842 on GOOGL and AMZN and 0.686 on AAPL, so this is systematic, not noise. A joint (start, end) grid search finds no window that fits all three. The paper's prices came from Bloomberg / EODHD / FinnHub, not Yahoo. **This matters most: it puts a floor under every other discrepancy. A quantity requiring no model at all already fails to reproduce, so the gap in the TradingAgents row cannot be attributed to the LLM swap alone.**
The ARR% column is internally inconsistent	GOOGL shows CR 7.78% next to ARR 8.09%; inverting the standard annualisation gives N ≈ 0.96 years for a three-month backtest. Whatever ARR% is, it is not the formula in Appendix 8.2.2, so that column is not reproducible even given the returns.
The target numbers are not empirically plausible to begin with	AAPL at +26.62% with SR 8.21 and MDD 0.91% in a quarter when buy-and-hold lost 5.23%. The authors say so themselves in a footnote ("the highest Sharpe Ratio exceeds our expected empirical range ... We report results as they are in our experiments faithfully").

6.4 What the DeepSeek substitution can and cannot explain

Can explain: throughput (about 1 LLM calls and 5.5 min per cell, against the paper's "11 LLM calls & 20+ tool calls"), and the structured-output fallback described in §5 — both real quality-of-path differences that plausibly move decisions.

Cannot explain: the Buy & Hold discrepancy, which involves no model; the missing news and macro channels, which are vendor-side; and the ARR% inconsistency, which is arithmetic. The substitution is therefore one contributor among several, and not the binding one. Our numbers are reported as *our* numbers and are not claimed to be comparable to Table 1.

6.5 What I would ask the authors

- Which vendor and which price convention (close vs adjusted, timezone, inclusive/exclusive end date) produced Table 1?
- What exactly is ARR%, and over what N?
- How is a rating mapped to a position size? The paper shows long/short arrows but never the mapping, and Table 1 is unreachable without it.

- Is there a snapshot of the news and sentiment feeds for the backtest window, without which the multimodal claim cannot be tested at all?
- Are the Table 1 numbers a single run or an average over runs? Given the run-to-run variation we measure at temperature 0, a single-run figure could differ materially from a repeat of the same experiment.

7. Reproducing this study

```
git clone https://github.com/ike0504/TradingAgents.git
cd TradingAgents && python -m venv .venv && .venv/Scripts/activate
pip install -e .  &&  echo DEEPSEEK_API_KEY=... > .env

python -m repro.run_sweep --arm baseline --trial 1 --smoke   # one cell, wiring check
python -m repro.run_budget                                   # full paired grid, off-peak only
python -m repro.cells     --arm baseline --arm no_debate     # per-cell table
python -m repro.aggregate --arm baseline --arm no_debate     # mean +/- std over trials
python -m repro.compare                                      # our run vs Table 1
python -m repro.report                                       # regenerate this document
```

Everything the study adds lives in `repro/` and `REPRODUCIBILITY.md`. Results are written outside the repo (`../results/`) so sweeps never enter a commit. No key is committed; `.env` is gitignored.